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Aryabhatta Maths Olympiad (AMO)

Level 1 — Practice Paper 3

CLASS 7

Questions	45 multiple-choice questions
Marks	100
Time	60 minutes
Section I	40 questions × 2 marks = 80 marks
Section II	5 Star Questions × 4 marks = 20 marks
Negative marking	No negative marking
Calculator	Not allowed
Chapters	Number System · Operations · Fractions, Decimals & Percentages · Algebra · Geometry · Mensuration · Data Handling · Applied Mathematics · Mathematical Reasoning

Instructions to candidates

- This paper has 45 multiple-choice questions. Each question has four answer choices, of which exactly one is correct.
- Section I (Q1–Q40) carries 2 marks per question. Section II (Q41–Q45) carries 4 marks per question.
- There is no negative marking. Unanswered questions score zero.
- Calculators are not permitted. Rough work may be done on a separate sheet.
- Mark your answers on the answer sheet at the end of the paper.
- All figures are not necessarily drawn to scale unless stated.

For parents and teachers

Print or view this paper in colour. Pictures are part of the questions.

Parts follow the AMO syllabus chapters. Chapter 7 is not part of the Class 7 syllabus, so the chapter numbers shown in the Part headings have gaps.

The answer sheet and the answer key are at the end. Remove only the answer key before the student begins.

Name:

School:

Roll No.:

SECTION I 40 questions × 2 marks each**Part 1 — Number System (Syllabus Chapter 1)**

Q1. What number should go in the box $\square$ to make this statement true?

$$\left(-\frac{5}{6}\right) \times \square = 1$$

- (A) $\frac{11}{6}$ (B) $\frac{5}{6}$ (C) $\frac{6}{5}$ (D) $-\frac{6}{5}$

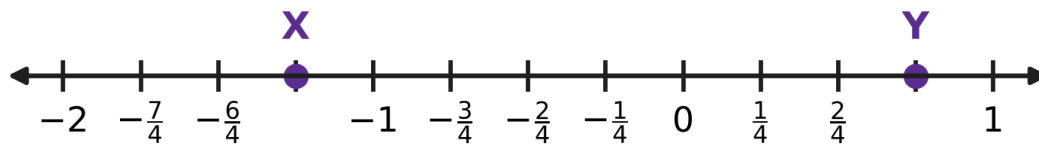
Q2. Which of these is NOT equal to $-\frac{5}{6}$?

- (A) $\frac{-10}{-12}$ (B) $\frac{-20}{24}$ (C) $\frac{15}{-18}$ (D) $\frac{-35}{42}$

Q3. What number should go in the box $\square$?

$$-\frac{2}{3} + \frac{5}{6} - \square = -\frac{1}{4}$$

- (A) $-\frac{1}{12}$ (B) $\frac{5}{12}$ (C) $-\frac{5}{12}$ (D) $\frac{1}{12}$



Q4. X and Y are marked on the number line. The gap between two whole numbers next to each other is split into quarters. Work out $X \div Y$.

- (A) $-\frac{5}{3}$ (B) $\frac{5}{3}$ (C) $-\frac{3}{5}$ (D) $-\frac{15}{16}$

Part 2 — Operations & Number Properties (Syllabus Chapter 2)

Q5. In the statement $7^4 \times 7^n \div 7^3 = 7^6$, the letter n stands for a missing whole number. What is n?

- (A) 9 (B) 2 (C) 5 (D) 13

Q6. What is the value of $(-2)^3 \times 3^0 + (-1)^5$?

- (A) 7 (B) -7 (C) -1 (D) -9

Q7. A large sorting centre of the post office handles 6×10^4 letters every day. How many letters does it handle in 30 days? Give the answer in standard form.

- (A) 1.8×10^4 letters (B) 1.8×10^5 letters
(C) 1.8×10^6 letters (D) 1.8×10^7 letters

Part 3 — Fractions, Decimals & Percentages (Syllabus Chapter 3)

Q8. The bus fare from a village to the town went up by 20%. The new fare is ₹42. What was the fare before the rise?

- (A) ₹22 (B) ₹35 (C) ₹33.60 (D) ₹50.40

Q9. What number should go in the box?

$$\square \% \text{ of } 80 = 30\% \text{ of } 120$$

- (A) 20 (B) 45 (C) 30 (D) 36

Q10. Hema makes clay pots. One pot costs her ₹160 to make. She sells it at a profit of 25%. What is the selling price of the pot?

- (A) ₹185 (B) ₹40 (C) ₹120 (D) ₹200

Q11. The school canteen sells a cup of milk for ₹20. Its offer is: "Buy 4 cups, get 1 cup free." Charu takes 5 cups under this offer. What percentage discount does she get on the marked price of the 5 cups?

- (A) 20% (B) 25% (C) 5% (D) 80%

Q12. A maths quiz has 60 questions. To win a badge, Juhi must get at least 70% of all the questions right. She has done the first 40 questions and got 29 of them right. What is the least number of the remaining 20 questions she must get right to win the badge?

- (A) 11 questions (B) 2 questions
(C) 13 questions (D) 14 questions

Part 4 — Algebra (Syllabus Chapter 4)

Q13. What is the value of $a^2b - 2ab^2$ when $a = -2$ and $b = 3$?

- (A) 48 (B) 24 (C) -48 (D) -24

Q14. The expression $6ab - 3a^2b + 2ba - a^2b + 7$ is simplified. What is the coefficient of a^2b in the simplified expression?

- (A) -3 (B) -2 (C) -4 (D) 8

Q15. The equation $4x - \square = 2x + 9$ has the solution $x = 7$. What number is in the box $\square$?

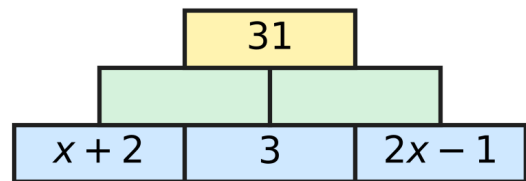
- (A) 23 (B) 5 (C) 19 (D) -5

Q16. Vikram's piggy bank has only ₹5 coins and ₹10 coins. There are 40 coins, and together they are worth ₹290. How many ₹10 coins are in the piggy bank?

- (A) 58 (B) 22 (C) 29 (D) 18

Q17. In the brick pyramid shown, the expression in each brick is the sum of the expressions in the two bricks just below it. The two middle bricks are left empty. The top brick is 31. What is the value of x ?

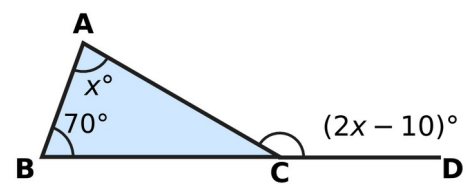
- (A) 9 (B) 8 (C) 24 (D) 10



Part 5 — Geometry (Syllabus Chapter 5)

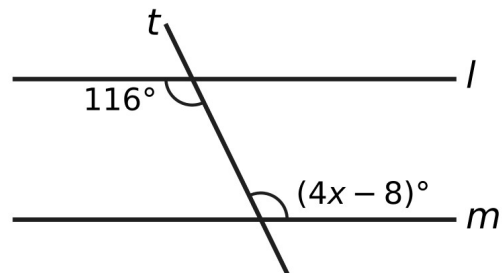
Q18. In the figure, side BC of $\triangle ABC$ is extended to D . $\angle BAC = x^\circ$, $\angle ABC = 70^\circ$ and the exterior angle $\angle ACD = (2x - 10)^\circ$. What is the value of x ?

- (A) 40 (B) 60 (C) 80 (D) 70



Q19. In the figure, the line t crosses the lines l and m . The two marked angles lie between l and m , on opposite sides of t . For which value of x will l be parallel to m ?

- (A) 29 (B) 18 (C) 27 (D) 31



Q20. Lalit proves that the angles of the quadrilateral $ABCD$ add up to 360° . The diagonal AC is drawn, and the angles 1, 2, 3, 4 are marked.

Step 1: AC divides $ABCD$ into $\triangle ABC$ and $\triangle ACD$.

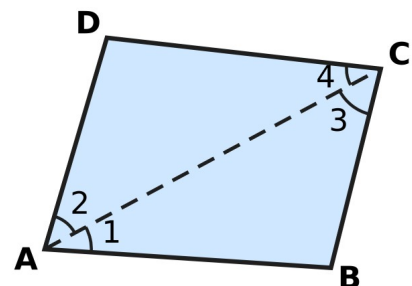
Step 2: $\angle 1 + \angle B + \angle 3 = 180^\circ$ and $\angle 2 + \angle D + \angle 4 = 180^\circ$.

Step 3: Adding, $\angle 1 + \angle 2 + \angle B + \angle 3 + \angle 4 + \angle D = 360^\circ$.

Step 4: $\angle 1 + \angle 2 = \angle A$ and $\angle 3 + \angle 4 = \angle C$, so $\angle A + \angle B + \angle C + \angle D = 360^\circ$.

Which of these facts does the proof NOT use?

- (A) Angles on a straight line add up to 180° .
 (B) The angles of a triangle add up to 180° .
 (C) A diagonal splits a quadrilateral into two triangles.
 (D) An angle split into two parts equals the sum of the parts.

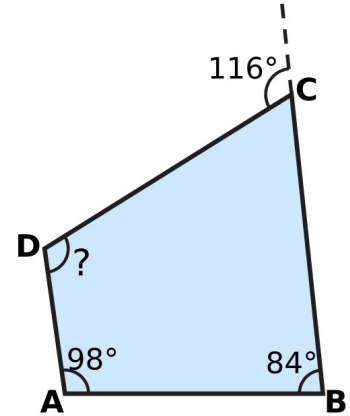


Q21. It is given that $\triangle PQR \cong \triangle LMN$. In $\triangle PQR$, $\angle Q = 64^\circ$. In $\triangle LMN$, $\angle N = 51^\circ$. What is the size of $\angle L$?

- (A) 51° (B) 65° (C) 115° (D) 64°

Q22. In the quadrilateral ABCD shown, $\angle A = 98^\circ$ and $\angle B = 84^\circ$. Side BC is extended beyond C, and the angle between this extension and CD is 116° . What is the size of $\angle D$?

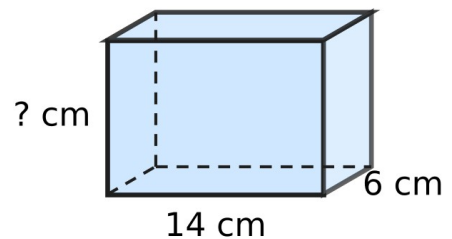
- (A) 114° (B) 62° (C) 178° (D) 66°



Part 6 — Mensuration (Syllabus Chapter 6)

Q23. The wooden block shown is a cuboid. Its volume is 840 cm^3 . It is 14 cm long and 6 cm wide. What is its height, marked "?"

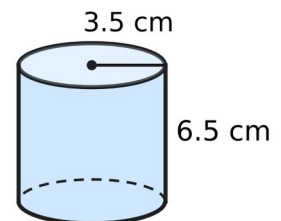
- (A) 140 cm (B) 42 cm (C) 60 cm (D) 10 cm



Q24. A closed tin is a cylinder of radius 3.5 cm and height 6.5 cm , as shown. How much metal sheet is needed to make it, including the top and the bottom?

Use $\pi = \frac{22}{7}$.

- (A) 181.5 cm^2 (B) 594 cm^2
(C) 220 cm^2 (D) 143 cm^2



Q25. A wooden cube of edge 16 cm is cut into small cubes, each of edge 4 cm . No wood is wasted. How many small cubes are made?

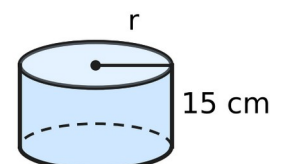
- (A) 16 (B) 64 (C) 4 (D) 1,024

Q26. Sahil cuts a wire of length 96 cm into 12 equal pieces and uses them as the 12 edges of a cube. He then covers all six faces of the cube with paper. What area of paper does he use?

- (A) 384 cm^2 (B) $1,536 \text{ cm}^2$
(C) 864 cm^2 (D) 256 cm^2

Q27. The curved surface area of the cylindrical container shown is $1,320 \text{ cm}^2$. Its height is 15 cm . What is its volume? Use $\pi = \frac{22}{7}$.

- (A) $18,480 \text{ cm}^3$ (B) $2,310 \text{ cm}^3$
(C) $9,240 \text{ cm}^3$ (D) $36,960 \text{ cm}^3$

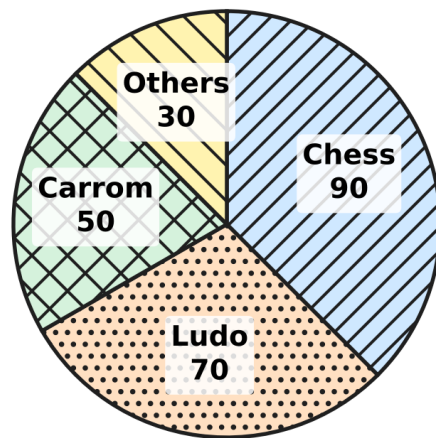


Part 7 — Data Handling, Statistics & Probability (Syllabus Chapter 8)

Q28. The pie chart shows how 240 students voted for their favourite board game. Each student voted once. What is the angle of the sector for Carrom?

- (A) 37.5° (B) 105° (C) 50° (D) 75°

Favourite board game



Total: 240 students

Q29. Sneha ran six laps of the school running track. Her lap times, in seconds, were 58, 64, 61, 55, 67 and 60. What is the median lap time?

- (A) 60.5 seconds (B) 58 seconds
(C) 60 seconds (D) 61 seconds

Q30. Avni counted the honey bees that visited each of five flower beds in one hour. The pictograph shows her counts. Half a symbol stands for 5 bees. What is the mean number of bees per flower bed?

- (A) 55 bees (B) 41 bees (C) 40 bees (D) 205 bees

Honey bees seen in one hour

Bed A 

Bed B 

Bed C 

Bed D 

Bed E 

 = 10 bees

 = half of that number

Part 8 — Applied Mathematics (Syllabus Chapter 9)

Q31. On Rhea's map, 2 cm stands for 15 km. Two towns are 7 cm apart on the map. What is the actual distance between them?

- (A) 30 km (B) 22 km (C) 105 km (D) 52.5 km

Q32. A pile of 15 identical textbooks weighs 9 kg. What do 25 of these textbooks weigh?

- (A) 225 kg (B) 5.4 kg (C) 15 kg (D) 19 kg

Q33. In a harbour, $\frac{3}{8}$ of the boats are fishing boats. All the other boats are sailing boats. What is the ratio of fishing boats to sailing boats?

- (A) 3 : 5 (B) 5 : 3 (C) 3 : 8 (D) 3 : 11

Q34. Ken repairs clocks in a clock shop. Inside one clock, a gear with 40 teeth turns a smaller gear with 24 teeth. The teeth of the two gears fit together, so the same number of teeth pass the meeting point on each gear. When the large gear makes 15 full turns, how many full turns does the small gear make?

- (A) 600 (B) 9 (C) 25 (D) 31

Folding 60 paper planes

Number of children	4	5	—
Planes folded by each child	15	—	6

Q35. Some children share the work of folding 60 paper planes equally. The table shows the number of planes each child folds for different numbers of children. Which numbers fill the two blanks? (Give the blank in the first row first.)

- (A) 12 and 10 (B) 10 and 14
(C) 13 and 14 (D) 10 and 12

Part 9 — Mathematical Reasoning & Problem-Solving (Syllabus Chapter 10)

Q36. The factors of 6 other than 6 itself are 1, 2 and 3, and $1 + 2 + 3 = 6$. Which of these numbers is also equal to the sum of its factors other than itself?

- (A) 20 (B) 28 (C) 24 (D) 12

Q37. Daksh has three number cards: 0, 4 and 7. He puts all three cards in a row to make a three-digit number. A three-digit number cannot start with 0. How many different three-digit numbers can he make?

- (A) 4 (B) 18 (C) 6 (D) 27

Q38. The five-digit number $4\square 152$ is divisible by 9. Which digit is in the box?

- (A) 7 (B) 1 (C) 3 (D) 6

Q39. Which of these numbers has 1 as its units digit?

- (A) 9^{35} (B) 7^{44} (C) 2^{40} (D) 3^{42}

Q40. The counting numbers from 1 onwards are added in order, and the sum is exactly 210. What is the last number added?

$$1 + 2 + 3 + \dots + \square = 210$$

- (A) 14 (B) 21 (C) 20 (D) 105

SECTION II — Star Questions 5 questions × 4 marks each

Q41. Some two-digit numbers are equal to the product of their two digits added to the sum of their two digits. For example, 25 does not have this property, because $2 \times 5 + 2 + 5 = 17$. How many two-digit numbers have this property? [4 marks]

- (A) 1 (B) 10 (C) 9 (D) 0

Q42. Nirav grows mangoes. It costs him ₹AB to grow 1 kg of mangoes, and he sells 1 kg at a profit of 20% for ₹BA. Here AB and BA are two-digit numbers made from two different non-zero digits A and B. What is the selling price of 1 kg? [4 marks]

- (A) ₹9 (B) ₹45 (C) ₹63 (D) ₹54

Q43. On one day, a museum sold 160 tickets. Adult tickets and child tickets were sold in the ratio 3 : 5. An adult ticket costs ₹20 more than a child ticket. The museum collected ₹6,000 from these tickets. What is the price of an adult ticket? [4 marks]

- (A) ₹50 (B) ₹30 (C) ₹45 (D) ₹25

Q44. A cuboid has a volume of 72 cm^3 . Its length, breadth and height are three different whole numbers of centimetres. A triangle can be made whose three sides have the same lengths as these three edges. What is the total surface area of the cuboid? [4 marks]

- (A) 124 cm^2 (B) 108 cm^2 (C) 178 cm^2 (D) 120 cm^2

Q45. The sum of two rational numbers is $-\frac{1}{6}$. One of them is $\frac{3}{4}$ more than twice the other. What is the larger of the two numbers? [4 marks]

- (A) $\frac{5}{36}$ (B) $\frac{41}{36}$ (C) $-\frac{11}{36}$ (D) $\frac{7}{36}$

Answer Sheet AMO Level 1 • Class 7 • Practice Paper 3**Name:****School:****Roll No.:***Shade or tick one circle for each question.*

- | | | |
|---------------------|---------------------|---------------------|
| 1. (A) (B) (C) (D) | 16. (A) (B) (C) (D) | 31. (A) (B) (C) (D) |
| 2. (A) (B) (C) (D) | 17. (A) (B) (C) (D) | 32. (A) (B) (C) (D) |
| 3. (A) (B) (C) (D) | 18. (A) (B) (C) (D) | 33. (A) (B) (C) (D) |
| 4. (A) (B) (C) (D) | 19. (A) (B) (C) (D) | 34. (A) (B) (C) (D) |
| 5. (A) (B) (C) (D) | 20. (A) (B) (C) (D) | 35. (A) (B) (C) (D) |
| 6. (A) (B) (C) (D) | 21. (A) (B) (C) (D) | 36. (A) (B) (C) (D) |
| 7. (A) (B) (C) (D) | 22. (A) (B) (C) (D) | 37. (A) (B) (C) (D) |
| 8. (A) (B) (C) (D) | 23. (A) (B) (C) (D) | 38. (A) (B) (C) (D) |
| 9. (A) (B) (C) (D) | 24. (A) (B) (C) (D) | 39. (A) (B) (C) (D) |
| 10. (A) (B) (C) (D) | 25. (A) (B) (C) (D) | 40. (A) (B) (C) (D) |
| 11. (A) (B) (C) (D) | 26. (A) (B) (C) (D) | 41. (A) (B) (C) (D) |
| 12. (A) (B) (C) (D) | 27. (A) (B) (C) (D) | 42. (A) (B) (C) (D) |
| 13. (A) (B) (C) (D) | 28. (A) (B) (C) (D) | 43. (A) (B) (C) (D) |
| 14. (A) (B) (C) (D) | 29. (A) (B) (C) (D) | 44. (A) (B) (C) (D) |
| 15. (A) (B) (C) (D) | 30. (A) (B) (C) (D) | 45. (A) (B) (C) (D) |

Q41–Q45 are Section II Star Questions (4 marks each).

For the parent or teacher

Section I correct: _____ / 40 × 2 = _____ marks

Section II correct: _____ / 5 × 4 = _____ marks

Correct: _____ / 45 **Score:** _____ / 100

Answer Key and Solutions AMO Level 1 • Class 7 • Practice Paper 3**Quick answers**

1 D	6 D	11 A	16 D	21 B	26 A	31 D	36 B	41 C
2 A	7 C	12 C	17 B	22 A	27 C	32 C	37 A	42 D
3 B	8 B	13 A	18 C	23 D	28 D	33 A	38 D	43 A
4 A	9 B	14 C	19 D	24 C	29 A	34 C	39 B	44 B
5 C	10 D	15 B	20 A	25 B	30 B	35 D	40 C	45 A

Solutions

Q	Part • level	Answer	Why?
1	Part 1 • Ch1 • Easy	(D) $-\frac{6}{5}$	A number times its reciprocal is 1. The reciprocal of $-5/6$ is $-6/5$, since $(-5/6) \times (-6/5) = 30/30 = 1$.
2	Part 1 • Ch1 • Easy	(A) $-\frac{10}{12}$	$-10/-12 = 10/12 = 5/6$, which is positive. The other three all simplify to $-5/6$.
3	Part 1 • Ch1 • Medium	(B) $\frac{5}{12}$	$-2/3 + 5/6 = -4/6 + 5/6 = 1/6$. Then $1/6 - \square = -1/4$, so $\square = 1/6 + 1/4 = 2/12 + 3/12 = 5/12$.
4	Part 1 • Ch1 • Medium	(A) $-\frac{5}{3}$	X is 5 quarter-steps left of 0, so $X = -5/4$. $Y = 3/4$. $X \div Y = (-5/4) \times (4/3) = -5/3$.
5	Part 2 • Ch2 • Easy	(C) 5	The left side is 7 to the power $4 + n - 3 = n + 1$. So $n + 1 = 6$ and $n = 5$.
6	Part 2 • Ch2 • Easy	(D) -9	$(-2)^3 = -8$ and $3^0 = 1$, so the product is -8. $(-1)^5 = -1$. The value is $-8 - 1 = -9$.
7	Part 2 • Ch2 • Medium	(C) 1.8×10^6 letters	$6 \times 10^4 \times 30 = 180 \times 10^4 = 1.8 \times 10^2 \times 10^4 = 1.8 \times 10^6$ letters (1,800,000).
8	Part 3 • Ch3 • Easy	(B) ₹35	The new fare is 120% of the old fare. So the old fare = $42 \times 100 \div 120 = ₹35$. Check: 20% of ₹35 is ₹7, and $₹35 + ₹7 = ₹42$.
9	Part 3 • Ch3 • Easy	(B) 45	30% of 120 = 36. 36 is $36/80 \times 100 = 45\%$ of 80.
10	Part 3 • Ch3 • Easy	(D) ₹200	Profit = 25% of ₹160 = ₹40. Selling price = $₹160 + ₹40 = ₹200$.
11	Part 3 • Ch3 • Medium	(A) 20%	Marked price of 5 cups = ₹100. She pays for 4 cups = ₹80. Discount = $₹20$ on $₹100 = 20\%$.
12	Part 3 • Ch3 • Medium	(C) 13 questions	70% of 60 = 42 right answers are needed. She already has 29, so she needs at least $42 - 29 = 13$ of the remaining 20.
13	Part 4 • Ch4 • Easy	(A) 48	$a^2b = 4 \times 3 = 12$. $2ab^2 = 2 \times (-2) \times 9 = -36$, so $-2ab^2 = +36$. The value is $12 + 36 = 48$.
14	Part 4 • Ch4 • Easy	(C) -4	The like terms in a^2b are $-3a^2b$ and $-a^2b$. Together they make $-4a^2b$, so the coefficient is -4.
15	Part 4 • Ch4 • Easy	(B) 5	Put $x = 7$: the left side is $28 - \square$ and the right side is $14 + 9 = 23$. So $\square = 28 - 23 = 5$.
16	Part 4 • Ch4 • Medium	(D) 18	Let the ₹10 coins be t . Then $10t + 5(40 - t) = 290$, so $5t + 200 = 290$ and $t = 18$.
17	Part 4 • Ch4 • Medium	(B) 8	Middle bricks: $(x + 2) + 3 = x + 5$ and $3 + (2x - 1) = 2x + 2$. Top: $3x + 7 = 31$, so $3x = 24$ and $x = 8$.

Q	Part • level	Answer	Why?
18	Part 5 • Ch5 • Easy	(C) 80	The exterior angle equals the sum of the two interior opposite angles: $2x - 10 = x + 70$, so $x = 80$. Check: $\angle ACD = 150^\circ = 80^\circ + 70^\circ$.
19	Part 5 • Ch5 • Easy	(D) 31	Between the lines and on opposite sides of t , the angles are alternate angles. The lines are parallel when they are equal: $4x - 8 = 116$, so $4x = 124$ and $x = 31$.
20	Part 5 • Ch5 • Easy	(A) Angles on a straight line add up to 180° .	Steps 1, 2 and 4 use the diagonal, the angle sum of a triangle and the parts of an angle. No straight-line angles appear anywhere in the proof.
21	Part 5 • Ch5 • Medium	(B) 65°	The order of the letters gives $P \leftrightarrow L$, $Q \leftrightarrow M$, $R \leftrightarrow N$, so $\angle R = \angle N = 51^\circ$. $\angle P = 180^\circ - 64^\circ - 51^\circ = 65^\circ$, and $\angle L = \angle P = 65^\circ$.
22	Part 5 • Ch5 • Medium	(A) 114°	Inside angle at $C = 180^\circ - 116^\circ = 64^\circ$. The angles of a quadrilateral add up to 360° , so $\angle D = 360^\circ - 98^\circ - 84^\circ - 64^\circ = 114^\circ$.
23	Part 6 • Ch6 • Easy	(D) 10 cm	Height = volume $\div$ (length $\times$ breadth) = $840 \div 84 = 10$ cm.
24	Part 6 • Ch6 • Easy	(C) 220 cm^2	Total surface area = $2\pi r(r + h) = 2 \times 22/7 \times 3.5 \times (3.5 + 6.5) = 22 \times 10 = 220 \text{ cm}^2$.
25	Part 6 • Ch6 • Easy	(B) 64	Volume of the big cube = $16 \times 16 \times 16 = 4,096 \text{ cm}^3$. Volume of a small cube = $4 \times 4 \times 4 = 64 \text{ cm}^3$. So $4,096 \div 64 = 64$ small cubes.
26	Part 6 • Ch6 • Medium	(A) 384 cm^2	Each edge = $96 \div 12 = 8$ cm. Each face = 64 cm^2 , and six faces = 384 cm^2 .
27	Part 6 • Ch6 • Medium	(C) $9,240 \text{ cm}^3$	$2\pi rh = 1,320$ gives $2 \times 22/7 \times r \times 15 = 1,320$, so $r = 14$ cm. Volume = $22/7 \times 14 \times 14 \times 15 = 9,240 \text{ cm}^3$.
28	Part 7 • Ch8 • Easy	(D) 75°	Angle = $50/240 \times 360^\circ = 75^\circ$.
29	Part 7 • Ch8 • Easy	(A) 60.5 seconds	In order: 55, 58, 60, 61, 64, 67. The two middle values are 60 and 61, so the median is $(60 + 61) \div 2 = 60.5$ seconds.
30	Part 7 • Ch8 • Medium	(B) 41 bees	The counts are 45, 30, 55, 40 and 35. Their total is 205, and $205 \div 5 = 41$ bees.
31	Part 8 • Ch9 • Easy	(D) 52.5 km	1 cm stands for $15 \div 2 = 7.5$ km. So 7 cm stands for $7 \times 7.5 = 52.5$ km.
32	Part 8 • Ch9 • Easy	(C) 15 kg	One textbook weighs $9 \div 15 = 0.6$ kg. So 25 textbooks weigh $25 \times 0.6 = 15$ kg.
33	Part 8 • Ch9 • Easy	(A) 3 : 5	If there are 8 parts in all, 3 parts are fishing boats and 5 parts are sailing boats. So the ratio is 3 : 5.
34	Part 8 • Ch9 • Medium	(C) 25	Teeth $\times$ turns is the same for both gears: $40 \times 15 = 600 = 24 \times$ turns. So the small gear makes $600 \div 24 = 25$ turns.
35	Part 8 • Ch9 • Medium	(D) 10 and 12	The number of children $\times$ planes per child is always $4 \times 15 = 60$. So $60 \div 6 = 10$ children, and $60 \div 5 = 12$ planes each.
36	Part 9 • Ch10 • Easy	(B) 28	For 28: $1 + 2 + 4 + 7 + 14 = 28$. For 12 the sum is 16, for 20 it is 22 and for 24 it is 36.
37	Part 9 • Ch10 • Easy	(A) 4	With 4 first: 407, 470. With 7 first: 704, 740. So there are 4 numbers.
38	Part 9 • Ch10 • Easy	(D) 6	$4 + 1 + 5 + 2 = 12$. The digit sum must be a multiple of 9, so $12 + \square = 18$ and $\square = 6$.
39	Part 9 • Ch10 • Medium	(B) 7^{44}	Units digits of powers of 7 repeat 7, 9, 3, 1. 44 is a multiple of 4, so 7^{44} ends in 1. 3^{42} ends in 9, 9^{35} ends in 9 and 2^{40} ends in 6.
40	Part 9 • Ch10 • Medium	(C) 20	$1 + 2 + \dots + n = n \times (n + 1) \div 2$. We need $n \times (n + 1) = 420 = 20 \times 21$, so the last number is 20.

Q	Part · level	Answer	Why?
41	Section II · Ch4 + Ch10 · Hard	(C) 9	Write the number as $10a + b$. Then $10a + b = ab + a + b$ gives $9a = ab$, so $b = 9$ (a is not 0). The numbers are 19, 29, ..., 99: there are 9.
42	Section II · Ch3 + Ch10 · Hard	(D) ₹54	120% of $AB = BA$, so $120(10A + B) = 100(10B + A)$. This gives $1,100A = 880B$, that is $5A = 4B$. With single digits, $A = 4$ and $B = 5$. Cost ₹45, selling price ₹54 (20% of ₹45 is ₹9).
43	Section II · Ch4 + Ch9 · Hard	(A) ₹50	Adult tickets = $3/8$ of $160 = 60$ and child tickets = 100. With child price c : $60(c + 20) + 100c = 6,000$, so $160c = 4,800$ and $c = 30$. Adult price = ₹50.
44	Section II · Ch5 + Ch6 · Hard	(B) 108 cm^2	Three different whole numbers with product 72: (1, 2, 36), (1, 3, 24), (1, 4, 18), (1, 6, 12), (1, 8, 9), (2, 3, 12), (2, 4, 9), (3, 4, 6). Only 3, 4, 6 make a triangle ($3 + 4 > 6$). $TSA = 2(12 + 24 + 18) = 108 \text{ cm}^2$.
45	Section II · Ch1 + Ch4 · Hard	(A) $\frac{5}{36}$	Let the other number be x . Then $x + (2x + 3/4) = -1/6$, so $3x = -1/6 - 3/4 = -11/12$ and $x = -11/36$. The first is $-22/36 + 27/36 = 5/36$, which is the larger.